Two matter insertions in a common moving chord frame

Additive null candidates from the ordered chord inner product

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Abstract

We connect the preceding moving-core limit of the Lin–Stanford generators to an ordered two-matter space with its actual chord inner product. A triangular oscillator realization provides a common background factor and two matter factors. Its finite-level Gram form and outer ladder action agree exactly with the chord rules. Displacing only the common factor gives convergence of every fixed finite generator word to the sum representation on the two matter factors. In particular, the limiting positive null candidate is additive. Positive commuting candidates can also be transported to the finite-deformation ordered space; their boost covariance holds in the specified moving limit. The underlying quantum-group factorization is established prior work. The result does not fix an opposite light-cone momentum, particle masses, or the identical-matter exchange completion.

1 What is being connected

Take the fixed-λ, half-filled double-scaling limit first, then $q = e^{-\epsilon} \to 1$, $\epsilon = 4\lambda$, with $K = H/(2J_s) = h/\sqrt{\epsilon}$ and $\hbar = 1$. We use the ordered direct two-matter module with weights $d_1, d_2 > 0$; the charged singlet example has $d_1 = d_2 = 2$. The order is retained. In particular, its norm is not the direct-plus-crossed norm of two identical insertions.

The project already resolved the two-matter primary channels and their standard Jacobi–Laguerre coupling on 6 November. We do not repeat that result as a new particle construction. The missing connection was between the actual large-background LS generators and the individual matter factors.

Lin–Stanford supply the ordered ladder rules [1]. Van der Heijden, Verlinde and Xu (HVX) construct the quantum-group factorization and its fusion/ordering operators [2]. Their deformation parameter obeys $q_{\mathrm{HVX}}^2 = q$ in our notation. The construction below is a chord-number realization of that established factorization, with the background at the left. Its moving-core estimate is the additional step needed here. We derive the ladder realization directly, avoiding dependence on normalization conventions in energy-space intertwiner formulas.

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2 An exact triangular realization

On $\ell^2(\mathbb{N}_0)^{\otimes 3}$ use coordinates (n, k_1, k_2). Let a act on the first factor with $a|n\rangle = \sqrt{[n]}|n-1\rangle$, where $[z] = (1-q^z)/(1-q)$. On matter factor i define

$$R_i = q^{d_i+N_i}, \qquad b_i|k_i\rangle = \sqrt{[k_i](1-q^{2d_i+k_i-1})}|k_i-1\rangle. \tag{1}$$

Define successively

$$A_0 = a, \qquad A_1 = A_0R_1 + b_1, \qquad A_2 = A_1R_2 + b_2 = aR_1R_2 + b_1R_2 + b_2. \tag{2}$$

Different tensor factors commute. All A_i are bounded at fixed q, and $A_iA_i^\dagger - qA_i^\dagger A_i = 1$. For example, $b_ib_i^\dagger - qb_i^\dagger b_i = 1 - R_i^2$; $R_ib_i^\dagger = qb_i^\dagger R_i$ cancels the cross terms. The three creation operators commute.

Proposition 1. *There is a unitary $\mathcal{F}_{q;d_1,d_2}$ from this tensor space to the ordered chord module such that*

$$a_L = A_0, \quad a_R = A_2, \quad \bar{n} = N_0 + N_1 + N_2 + d_1 + d_2. \tag{3}$$

Explicitly, the raw chord state $|l, m, r\rangle$ corresponds to $(A_0^\dagger)^l(A_1^\dagger)^m(A_2^\dagger)^r|0, 0, 0\rangle$.

Proof. Commuting the outer annihilators through this product yields

$$\begin{aligned} a_L|l, m, r\rangle = [l]|l-1, m, r\rangle + q^{d_1+l}[m]|l, m-1, r\rangle \\ + q^{d_1+d_2+l+m}[r]|l, m, r-1\rangle, \end{aligned} \tag{4}$$

and the reflected right action. These are the ordered chord rules. For the left action use $[a, A_1^\dagger] = q^{N_0+d_1+N_1}$ and $[a, A_2^\dagger] = q^{\bar{n}}$. For the right action first remove the right creators using its q-oscillator relation; on the subspace with $k_2 = 0$, the remaining A_2 is $q^{d_2}A_1$. Removing the left creators uses $[A_2, a^\dagger] = q^{\bar{n}}$.

Adjointness recursively removes every exterior chord from either entry of a Gram matrix element. The remaining pure-middle norm is $[m]!$, since A_1 is a q-oscillator annihilating the vacuum. Thus the entire direct Gram form agrees, not only the outer algebra. At each total grade, the creation map is triangular: its coefficient at $(n, k_1, k_2) = (l, m, r)$ is nonzero, and no term has $k_2 > r$, or $k_2 = r, k_1 > m$. It is invertible at every grade and extends to the claimed unitary. □

3 The joint moving-core limit

Let $D(-\sqrt{m})$ be the ordinary oscillator displacement on the common factor, and set

$$\mathcal{V} = \mathcal{F}_{q;d_1,d_2}(D(-\sqrt{m}) \otimes 1 \otimes 1). \tag{5}$$

Here m is a displacement parameter, not the middle raw chord number in (4). The core consists of finite linear combinations of $|j, k_1, k_2\rangle$. On factor i define

$$L_{i,-}|k\rangle = \sqrt{k(2d_i + k - 1)}|k-1\rangle, \quad E_i = d_i + N_i, \quad B_i = (L_{i,+} + L_{i,-})/2, \quad P_i = i(L_{i,+} - L_{i,-})/2.$$

Write $E_\Sigma = E_1 + E_2$, $B_\Sigma = B_1 + B_2$, $P_\Sigma = P_1 + P_2$ and $T_\Sigma = -B_\Sigma$.

Proposition 2. *If $m \to \infty$ and $\epsilon m \to 0$, every fixed finite noncommutative polynomial p satisfies, for each core vector v,*

$$\|p(E, \widehat{B}, \widehat{P}, B_{\mathrm{tr}})\mathcal{V}v - \mathcal{V}p(E_\Sigma, B_\Sigma, P_\Sigma, T_\Sigma)v\| \to 0. \tag{6}$$

The LS conventions are those of the preceding note: $E = ([\bar{n}] - (a_L^\dagger a_L + a_R^\dagger a_R)/2)/c$, $\widehat{B} = (a_L^\dagger a_L - a_R^\dagger a_R)/(2c\sqrt{1-c^2})$, $\widehat{P} = i(a_L^\dagger a_R - a_R^\dagger a_L)/(2\sqrt{1-c^2})$, $c = q^{\bar{n}/2}$, and $B_{\mathrm{tr}} = (K_L - K_R)/2$.

Proof. Put $w = d_1 + d_2 + N_1 + N_2$, $R = R_1R_2 = q^w$, $C = b_1R_2 + b_2$, so $a_R = aR + C$. The exact stable expressions are

$$\begin{aligned} Z &= a^\dagger RC + C^\dagger Ra + C^\dagger C, \\ E &= \frac{[w] - (1-R)^2[N_0]/2 - Z/2}{c}, \\ \widehat{B} &= \frac{(1-R^2)[N_0] - Z}{2c\sqrt{1-c^2}}, \qquad \widehat{P} = \frac{i(a^\dagger C - C^\dagger a)}{2\sqrt{1-c^2}}, \end{aligned} \tag{7}$$

$$B_{\mathrm{tr}} = \frac{(a + a^\dagger)(1-R) - (C + C^\dagger)}{2\sqrt{\epsilon}}. \tag{8}$$

Every numerator preserves the total grade where the displayed denominator is used. For bounded k_1, k_2, $C/\sqrt{\epsilon} \to L_{1,-} + L_{2,-}$, while $R_i \to 1$. On $m/2 \le n \le 2m$, the common diagonal terms of $\widehat{B}$ and the common-oscillator part of (8) are $O(\sqrt{\epsilon m})$. The E cross terms have the same order, and $E \to w$. The $C^\dagger C$ contribution to $\widehat{B}$ is $O(\sqrt{\epsilon/m})$ there. The common shifts accompanying C in $\widehat{B}, \widehat{P}$ act as -1 on displaced fixed descendants, giving B_Σ, P_Σ with the displayed signs. □

4 Tails, positive candidates and the meaning of addition

To finish the proof of Proposition 2, write $\zeta_j = D(-\sqrt{m})|j\rangle$. Its coefficients are a fixed-degree Charlier polynomial times the Poisson amplitude, and

$$\|(N_0/m - 1)\zeta_j\|^2 = (2j+1)/m + j^2/m^2.$$

As in the preceding one-matter proof, both common unilateral shifts approach -1 on these vectors. A word of length L changes the internal levels and n by bounded amounts. From (7) its finite-path coefficients admit an envelope $C(1+\epsilon(n+L))^L \exp(C_L\epsilon(n+w+L))$ for bounded initial internal levels. Squared envelopes times the polynomial-Poisson weight have exponentially small tails outside $[m/2, 2m]$: tilting changes the Poisson mean to $me^{O_L(\epsilon)}$ and its prefactor to $\exp(m(e^{O_L(\epsilon)} - 1)) \to 1$. The bulk estimates and finite path expansion prove (6) for products directly. They also apply if a fixed number of the limiting internal generators is inserted between actual generators.

Define $Q_i = E_i - P_i$. Each has the positive form

$$\langle v, Q_i v\rangle = \tfrac{1}{2}\sum_{k\geq 0} |\sqrt{2d_i + k}\, v_k + i\sqrt{k+1}\, v_{k+1}|^2.$$

Its Jacobi closure is self-adjoint and positive; different factors strongly commute. Consequently

$$Q_\Sigma = Q_1 + Q_2 \geq 0, \qquad [T_\Sigma, Q_i] = iQ_i, \qquad E - \widehat{P} \;\longrightarrow\; Q_\Sigma. \tag{9}$$

The last arrow has the moving-core meaning of (6). This is addition in an explicit microscopic ordered-space identification, not an assumption about the bare concatenation coproduct. The common oscillator supplies a multiplicity factor.

There are also positive, strongly commuting operators on the finite-q ordered space:

$$\mathfrak{Q}_i^{(q)} = \mathcal{F}_{q;d_1,d_2}(1 \otimes Q_i)\mathcal{F}_{q;d_1,d_2}^\dagger. \tag{10}$$

The notation $1 \otimes Q_i$ means the appropriate matter factor. Because these commute with the common displacement, the mixed-word estimate gives

$$\|([B_{\rm tr}, \mathfrak{Q}_i^{(q)}] - i\mathfrak{Q}_i^{(q)})\mathcal{V}v\| \to 0, \qquad \|(E - \widehat{P} - \mathfrak{Q}_1^{(q)} - \mathfrak{Q}_2^{(q)})\mathcal{V}v\| \to 0.$$

The transported operators are specified candidates, not a uniqueness statement or single-sided local observables. The finite-q LS combination $E - \widehat{P}$ itself need not be positive. The limit operators satisfy the usual Carleman core condition; the conjugated individual self-adjoint LS operators therefore also converge in the strong-resolvent sense.

5 Exchange, the other null direction, and mass

The ordered direct space must be distinguished from the physical two-open-X Wick sector for identical singlets. The latter adds a crossed Gram form. Our earlier first-channel calculation gives, at $d_1 = d_2 = 2$,

$$\|\Omega_0\|^2_{\text{phys}} = 1 + q^4, \qquad \|\Omega_1\|^2_{\text{phys}} = 1 - q^8 \sim 8\epsilon, \tag{11}$$

where the primaries are unit normalized in the direct form. The crossed overlap intertwines the outer chord action, so the same factors multiply all descendants in these two channels. Neither displacement nor a large number of common chords removes this mismatch. Uniformly importing individual-factor operators into the identical-matter completion requires the crossed metric, including its singular normalization.

There is useful newer literature here. The LS general channel overlap was originally supported by a finite-level conjecture; HVX derive the corresponding fusion-channel swapping weight from an Askey–Wilson transform [2], (82)–(83). This updates the project's literature map. It does not by itself identify swapping with a unitary scattering matrix or resolve our identical-matter moving limit. We leave the older sealed calculation intact and use the newer result as a subsequent input.

Even after ordered addition, an opposite momentum is not fixed. The spectral representation of Q_Σ, T_Σ can be written as $L^2(\mathbb{R}_+, dP) \otimes \mathcal{K}$, with $Q_\Sigma = P$ and $T_\Sigma = i(P\partial_P + 1/2)$. Here $\mathcal{K}$ includes the relative coordinate and, if retained, the common multiplicity. A positive self-adjoint operator strongly commuting with P and obeying the opposite dilation covariance is of the form

$$P^- = \frac{A}{\mu P}, \qquad P^+ = \mu P, \qquad A \geq 0 \text{ on } \mathcal{K}. \tag{12}$$

Indeed strong commutation makes it a measurable operator field in P; dilation covariance makes $PP^-(P)$ constant, up to null sets. Equivalently the logarithmic variable makes this a translation-invariant field. Appropriate self-adjoint domains are understood through the joint spectral calculus. Thus the internal operator $A = P^+P^-$ remains free kinematic data. For two freely assigned masses its usual choice is $A = m_1^2/x + m_2^2/(1-x)$, $x = p_1/(p_1+p_2)$; the chord boost and positive Q alone do not select it.

The other LS limit $S = E_\Sigma + P_\Sigma$ obeys $[Q_\Sigma, S] = -2iT_\Sigma$, and hence is not a commuting opposite Poincare momentum. This observation does not require global de Sitter translations to commute; it concerns the specific flat-space particle interpretation sought here. Selected-state ordering, four-bilocal dynamics and finite-G effects must still supply additional physical information.

6 Verification and next target

Independent rational Gram recursion and the triangular images agree in 1,484 entries, with 280 outer lowering actions checked. Parameters are $q = 1/4, 4/9$, $(d_1, d_2) = (2,2), (1,2)$ and total grades zero through four. Monic orthogonal coordinates make these comparisons rational and avoid numerical Gram degeneracy.

The moving check uses $m = 8, 32, 128, 512$, $\epsilon = m^{-3}$, $(d_1, d_2) = (2,2)$, and $(j, k_1, k_2) = (0,0,0), (1,1,0), (2,1,2)$. Eight words and both $[\widehat{B}, \widehat{P}] - iE$ and $[B_{\mathrm{tr}}, Q] - iQ$ are tested, with $Q = E - \widehat{P}$. Common cutoffs differ by 48, and all axes are padded beyond the word shifts. The input tail is bounded by a polynomial-weighted Poisson Chernoff estimate; it does not on its own certify unbounded-word errors.

m	relative Q error	$\|([B_{\mathrm{tr}}, Q] - iQ)\mathcal{V}\|2,1,2\rangle\|$
8	0.200467	3.222535
32	0.090199	1.181758
128	0.042464	0.414286
512	0.020876	0.173320

Here the relative Q error uses $\|Q_\Sigma|2,1,2\rangle\|$ as denominator. At $m = 128$ the same nontrivial descendant was independently evaluated with 60-decimal sparse outer-ladder composition. This uses the direct LS definition, whereas the double-precision code uses (7). Eight quantities were compared, including the two commutator defects themselves; the largest relative vector discrepancy was 5.93×10^{-13}. Full cutoffs, tail bounds, cancellation ratios, versions and input/output hashes accompany the data. Refinement and high-precision agreement are diagnostics, not interval arithmetic certificates.

The next target is the crossed metric in this explicit moving frame, followed by its effect on the actual identical-singlet four-point function. This preserves the singular first relative channel rather than discarding it as a free-limit null vector. The opposite momentum and mass require that dynamical step; repeating the standard tensor decomposition will not determine them.

References

[1] H. W. Lin and D. Stanford, *A symmetry algebra in double-scaled SYK*, arXiv:2307.15725, Secs. 2.3, 3.3 and Apps. B.2, F.

[2] J. van der Heijden, E. Verlinde and J. Xu, *Quantum Symmetry and Geometry in Double-Scaled SYK*, arXiv:2511.08743v2, Secs. 3.1–3.4, especially (55), (62), (82)–(83), (92).